Advanced Predictive Modeling and Adaptive Algorithms for Personalized Postprandial Glucose Forecasting in Digital Health Applications

1. Introduction

The regulation of postprandial blood glucose (BG) remains one of the most critical and complex challenges in the management of diabetes mellitus. Historically, insulin dosing, glycemic forecasting, and medical nutrition therapy have relied almost exclusively on carbohydrate quantification. However, this one-dimensional approach fails to account for the profound metabolic impact of other macronutrients, such as dietary fats and proteins, which significantly alter gastric emptying rates, hepatic glucose production, and insulin sensitivity.[1] As continuous glucose monitoring (CGM) technologies, wearable devices, and digital health platforms advance, the paradigm of diabetes management is shifting toward Software as a Medical Device (SaMD) solutions that integrate complex multivariable physiological interactions.[2]

The development of a personalized digital health application capable of projecting meal-induced glucose excursions and dynamically refining its predictions based on subsequent physiological feedback represents a frontier in medical informatics. By acquiring an initial pre-meal blood glucose reading, logging the exact macronutrient composition of the intended meal, and subsequently feeding a mid-course postprandial reading (e.g., 30 minutes post-ingestion) back into the system, an adaptive algorithm can continuously recalibrate the patient's metabolic profile.[2] This recursively updating architecture operates on two distinct temporal horizons: it provides immediate predictive course correction for the active meal using state-estimation techniques, and it longitudinally personalizes the patient's Insulin-to-Carbohydrate Ratio (ICR) and Insulin Sensitivity Factor (ISF) through machine learning and run-to-run (R2R) control mechanisms.[4]

The objective of this comprehensive report is to provide an exhaustive technical and clinical framework for engineering such an application. It delineates medically approved heuristic formulas for mixed-meal glucose impact, extensively details the physiological differential equations governing glucose-insulin dynamics, outlines the mathematical architecture of recursive adaptive algorithms for real-time prediction updates, and rigorously reviews the United States Food and Drug Administration (FDA) regulatory requirements for deploying predictive insulin-adjustment software under Product Code QRX.

2. The Physiological Impact of Mixed Macronutrients on Glycemic Dynamics

While carbohydrates are the primary and most immediate drivers of postprandial blood glucose spikes, clinical research unequivocally demonstrates that the presence of fat and protein in a meal exerts a profound, delayed, and additive effect on glycemic excursions.[1] Designing a predictive application necessitates a deep mathematical and physiological understanding of how these macronutrients behave in vivo.

2.1 Carbohydrate Kinetics

Dietary carbohydrates (CHOs) are the principal source of postprandial blood glucose. Once ingested, polysaccharides and disaccharides are enzymatically cleaved into monosaccharides (primarily glucose) in the small intestine, leading to rapid absorption into the portal circulation.[7]

The rate of glucose appearance (R_a) from pure carbohydrates typically peaks between 45 and 90 minutes post-ingestion.[9] However, the bioavailability and absorption rate of carbohydrates are heavily modulated by the physical matrix of the food, particularly the presence of dietary fiber. Fiber resists enzymatic digestion in the human small intestine, thereby slowing gastric emptying and dampening the postprandial glycemic peak. In standard clinical practice, and within application logic, available (or "net") carbohydrates are calculated by subtracting total dietary fiber from total carbohydrates.[10]

2.2 The Glycemic Impact of Dietary Fat

Dietary fat does not directly convert to glucose; however, it profoundly alters glucose pharmacodynamics. High-fat meals (typically defined as containing >30 to 40 grams of fat) initiate a cascade of physiological responses that complicate insulin dosing.[1]

Firstly, fat triggers the release of gut hormones such as cholecystokinin (CCK) and glucagon-like peptide-1 (GLP-1), which significantly delay gastric emptying.[1] Consequently, the absorption of co-ingested carbohydrates is protracted, often leading to a flatter but much wider postprandial glucose curve. If a patient administers a standard rapid-acting insulin bolus for a high-fat meal, the peak action of the insulin may occur before the bulk of the carbohydrates are absorbed, precipitating early postprandial hypoglycemia (within the first 1 to 2 hours).[1]

Secondly, dietary fat and the subsequent elevation of circulating free fatty acids (FFAs) acutely impair cellular insulin sensitivity and enhance hepatic glucose production (gluconeogenesis).[1] Clinical studies, such as those conducted by Wolpert et al., demonstrate that the consumption of 50 grams of fat can effectively double the postprandial insulin requirement, inducing late-onset hyperglycemia that persists for 3 to 6 hours post-ingestion.[6]

2.3 The Glycemic Impact of Dietary Protein

Similar to fat, dietary protein exerts an additive and delayed effect on blood glucose levels.[6] While amino acids derived from protein digestion can be utilized for hepatic gluconeogenesis, the primary glycemic impact of protein relates to its endocrine interactions.[1] High-protein meals stimulate the secretion of glucagon, a hormone that counteracts insulin by promoting glycogenolysis and gluconeogenesis in the liver.[1]

Clinical trials indicate that meals containing more than 25 grams of protein, when consumed with carbohydrates, significantly increase the total insulin requirement and contribute to delayed glucose excursions.[11] For individuals on low-carbohydrate diets, the gluconeogenic conversion of protein becomes a primary source of circulating glucose, necessitating explicit algorithmic accounting within a predictive application.[13]

2.4 Modulating Factors: Caffeine and Alcohol

Beyond standard macronutrients, a comprehensive predictive algorithm should eventually account for other common dietary variables such as caffeine and alcohol. Caffeine acts as an adenosine receptor antagonist and stimulates the release of epinephrine, which can acutely reduce insulin sensitivity and elevate fasting and postprandial glucose levels. Clinical observations note that intakes exceeding 200 milligrams of caffeine can cause noticeable hyperglycemic shifts in individuals with diabetes.[14] Conversely, alcohol inhibits hepatic gluconeogenesis, presenting a significant risk for delayed, severe hypoglycemia, particularly when consumed without concomitant carbohydrates.[16]

3. Medically Approved Heuristics and Formulas for Mixed-Meal Insulin Dosing

To algorithmically project the glycemic impact of a logged meal before dynamic sensor feedback is available, the application must translate the raw macronutrient data (grams of carbs, proteins, and fats) into a predicted insulin demand. Several clinically validated heuristic formulas have been developed by the medical community to quantify this demand. These formulas provide the foundational logic for an application's initial predictive curve.

3.1 The Total Available Glucose (TAG) Model

The Total Available Glucose (TAG) methodology is one of the earliest systematic approaches to accounting for mixed macronutrients. It operates on the physiological premise that different macronutrients yield glucose at varying rates and conversion efficiencies.[18]

The TAG algorithm calculates the total absolute glycemic load of a meal using a standardized mass-to-glucose conversion percentage. The clinical consensus parameters for the TAG formula dictate that 100% of digestible carbohydrates, approximately 58% of proteins, and 10%

of fats are eventually metabolized into glucose.[18]

The algorithmic equation utilized within a digital framework is structured as:

$$TAG(g) = Carbohydrates(g) + (0.58 \times Protein(g)) + (0.10 \times Fat(g))$$

While the TAG method provides a unified numerical output that allows for precise, albeit static, insulin adjustments, it possesses inherent limitations for continuous forecasting.[18] Specifically, the TAG formula aggregates the glycemic potential but does not temporally model the physiological delay caused by fat and protein digestion. Therefore, while it accurately predicts the *total* insulin required, it fails to dictate the *timing* of the delivery, rendering it insufficient for advanced closed-loop or predictive applications without additional temporal modifiers.

3.2 The Warsaw Method and the Pankowska Equation

To address the temporal limitations of early formulas, the Warsaw Pump Therapy School developed the Pankowska Equation. This methodology provides a highly sophisticated framework for mixed-meal management by introducing the concept of Fat-Protein Units (FPUs) and formally linking macronutrient load to extended insulin delivery protocols.[20]

An FPU is defined strictly as 100 kilocalories (kcal) derived entirely from dietary fat and/or protein.[21] The clinical heuristic established by Pankowska dictates that the metabolic impact of one FPU requires the exact same amount of insulin as 10 grams of carbohydrates (which is defined as one Carbohydrate Unit, or CU).[21]

The step-by-step algorithmic calculation for software implementation is as follows:

1. **Calculate Non-Carbohydrate Caloric Load:**
   $Calories_{Fat/Pro} = (Fat(g) \times 9 \text{ kcal/g}) + (Protein(g) \times 4 \text{ kcal/g})$ [22]
2. **Determine the Number of FPUs:** $FPU = \frac{Calories_{Fat/Pro}}{100}$ [23]
3. **Calculate the Extended Insulin Dose:** The additional insulin required to cover the FPU load is derived by dividing the FPU's carbohydrate equivalent (FPU $\times$ 10g) by the patient's personalized Insulin-to-Carbohydrate Ratio (ICR).[22] $Insulin_{FPU} = \frac{FPU \times 10}{ICR}$ [25]
4. **Temporal Distribution (The Extended Wave):**
   Because fat and protein profoundly delay absorption, the Pankowska method mandates that the insulin calculated for FPUs must not be delivered immediately. Instead, it must be programmed as an extended (square-wave) bolus over a duration that is directly proportional to the total number of FPUs:
   - 1 FPU $\rightarrow$ Extended over 3 hours
   - 2 FPUs $\rightarrow$ Extended over 4 hours

- 3 FPUs $\rightarrow$ Extended over 5 hours
- 3 FPUs $\rightarrow$ Extended over 8 hours.[22]

Clinical Validation and Algorithmic Limitations: While the Pankowska Equation is highly effective at neutralizing late postprandial hyperglycemia (hours 3 to 6), rigorous randomized crossover trials have highlighted a significant clinical risk. By aggressively increasing the total insulin volume, the Pankowska method frequently induces early postprandial hypoglycemia (hours 1 to 2).[21] Consequently, if this formula is hardcoded into a predictive application, it must be paired with dynamic safety constraints that can independently suspend basal insulin if a 30-minute post-meal reading trends downward.

3.3 The Sieradzki Equation

To mitigate the acute hypoglycemic risks associated with the Pankowska Equation and to reduce the computational and user-input burden, the Sieradzki Equation (SE) was developed as a simplified, yet highly effective, alternative.[21] Rather than calculating distinct caloric values for fats and proteins, the Sieradzki Equation applies a standardized proportional increase to the baseline carbohydrate bolus when a meal exceeds specific fat and protein thresholds (e.g., >40g fat and >25g protein).[21]

The Sieradzki algorithmic protocol operates as follows:

1. **Standard Bolus Delivery:** Calculate the insulin required solely for the carbohydrate content ($Insulin_{CHO} = \frac{Carbohydrates(g)}{ICR}$) and deliver it immediately prior to the meal.[21]
2. **Extended Bolus Calculation:** Calculate an additional 30% of the initial carbohydrate insulin dose to cover the fat and protein load. $Insulin_{Extended} = 0.30 \times Insulin_{CHO}$ [29]
3. **Temporal Distribution:** Deliver this extended dose uniformly over a fixed 4-hour window.[21]

Comparative Clinical Efficacy: Recent clinical evaluations, including double-blind randomized crossover trials comparing the Pankowska and Sieradzki methods, demonstrate that while both algorithms effectively maintain postprandial glucose within the broad target range (70–180 mg/dL), the Sieradzki Equation is clinically superior regarding patient safety.[21] The Sieradzki protocol achieved a statistically longer Time in Tight Range (TITR: 70–140 mg/dL) at 82.51%, compared to 70.49% for the Pankowska method.[21] Crucially, the SE method resulted in significantly fewer hypoglycemic episodes at the 180-minute and 300-minute marks.[30] For software engineering, the Sieradzki Equation provides a safer, more stable baseline for generating initial predictive curves.

Clinical Metric	Pankowska Equation	Sieradzki Equation	Algorithmic Implication for SaMD
Input Complexity	High (Requires exact fat/protein grams to calculate kcal)	Low (Proportional scaling based on carbohydrate dose)	SE requires less friction in user interface logging.
Early Glycemia (60 min)	Higher mean glucose (136 mg/dL)	Lower mean glucose (124 mg/dL)	PE requires aggressive early basal suppression if the trend falls.
Late Hypoglycemia Risk	Higher risk due to strict FPU extended durations	Lower risk due to fixed 4-hour 30% distribution	SE provides a safer baseline prediction for machine learning models.
Time in Tight Range (TITR)	70.49%	82.51%	SE demonstrates superior stability in recent 2024/2025 trials.[21]

3.4 The Food Insulin Index (FII)

An alternative to strict macronutrient mathematical counting is the Food Insulin Index (FII). The FII is a physiological algorithm that ranks foods based on the actual in vivo insulin demand they evoke relative to an isoenergetic reference food (typically pure glucose).[31] The FII inherently captures the complex, synergistic metabolic effects of mixed macronutrients, including the dampening impact of dietary fiber and the unique insulinotropic properties of specific amino acids.[33]

Clinical trials evaluating the FII demonstrate significant reductions in the Mean Amplitude of Glycemic Excursions (MAGE) when compared to traditional carbohydrate counting.[31] However, the FII relies on empirically derived databases rather than real-time algorithmic inputs, limiting its utility for novel or unlisted mixed meals. For an advanced predictive application, the optimal architecture involves hybridizing an FII database lookup for known foods with the Sieradzki or Pankowska mathematical models for custom user-generated meals.

4. Physiological Mathematical Modeling of Glucose-Insulin Dynamics

While clinical heuristics (TAG, Sieradzki) provide static insulin dosing guidelines, an application designed to forecast a continuous, minute-by-minute blood glucose curve must utilize sophisticated compartmental differential equation models. These models translate the macronutrient inputs into dynamic state variables representing the gastrointestinal tract, blood plasma, and interstitial fluid.[34]

4.1 The Bergman Minimal Model

The foundational mathematical architecture for simulating in vivo diabetes dynamics is the Bergman Minimal Model.[35] It characterizes the rate of glucose appearance and disappearance through an elegant system of differential equations, balancing physiological accuracy with computational feasibility for mobile devices.

The glucose disappearance kinetics are defined by the equation:

$$\dot{G}(t) = -[p_1 + X(t)]G(t) + p_1 G_b + \frac{R_a(t)}{V_G}$$ [36]

Where:

- $G(t)$ represents the active plasma glucose concentration.
- G_b is the patient's basal (fasting) glucose level.
- $X(t)$ represents the insulin action state (the physiological effect of insulin on accelerating cellular glucose uptake).
- p_1 is the glucose effectiveness parameter (the rate of insulin-independent glucose utilization).
- V_G is the glucose distribution volume.
- $R_a(t)$ is the rate of glucose appearance in the blood resulting from the digested meal.[36]

The corresponding insulin action state, $X(t)$, is governed by:

$$\dot{X}(t) = -p_2 X(t) + p_3[I(t) - I_b]$$ [36]

Where $I(t)$ is the plasma insulin concentration, I_b is the basal insulin concentration, p_2 is the rate of decrease in insulin action, and p_3 dictates the insulin sensitivity index ($S_I = p_3/p_2$). The parameters $p_1, p_2,$ and p_3 are highly individualized. Because they fluctuate based on

circadian rhythms, physical activity, and stress, static assignments of these parameters will inevitably lead to predictive decay, necessitating the real-time feedback loops discussed in Section 5.[36]

4.2 The Hovorka Compartmental Model

To accurately forecast the glycemic impact of the mixed meals logged in the application, the glucose rate of appearance, $R_a(t)$, must be meticulously modeled. The Hovorka model significantly improves upon the Bergman architecture by incorporating an explicit gastrointestinal absorption sub-model, making it the gold standard for artificial pancreas research.[34]

In the Hovorka framework, gut absorption is represented by a two-compartment chain to accurately simulate the physiological delay caused by mechanical digestion and intestinal transport:

$$R_a(t) = \frac{A_G \cdot D_G \cdot t \cdot e^{-t/t_{max,G}}}{t_{max,G}^2}$$

Where:

- D_G is the total mass of carbohydrates consumed.
- A_G is the carbohydrate bioavailability factor (typically constrained around 0.8).[40]
- $t_{max,G}$ is the time-to-maximum absorption rate.[40]

Hovorka Model Parameter	Symbol	Standard Value	Physiological Definition
Insulin elimination	k_e	0.138 min^{-1}	Rate of insulin clearance from plasma.[41]
Insulin distribution	V_I	0.12 L/kg	Volume of insulin distribution.[41]
Glucose distribution	V_G	0.16 L/kg	Volume of glucose distribution.[41]
Carb	A_G	~0.8	Proportion of

bioavailability			ingested carbs entering blood.[41]

Algorithmic Integration of Macronutrient Effects into the Hovorka Model: To allow the application to dynamically project the effect of mixed meals, the absorption parameter $t_{max,G}$ must be algorithmically adjusted based on the fat and protein content logged by the user. If the user logs a High-Fat/High-Protein meal (e.g., pizza or steak), the application extends $t_{max,G}$ via a predefined, mathematically derived time shift: $\hat{t}_{max,G} = t_{max,G} + t_d$ [34] In this adaptation, t_d represents the physiological digestion delay induced by the fat and protein mass (e.g., an additional 20 to 60 minutes depending on the calculated FPU mass). This shift flattens the predictive curve, accurately modeling the delayed hyper-glycemic tail observed in clinical practice.[34]

4.3 The Dalla Man Model

The Dalla Man (DM) model further refines physiological predictions by simulating endogenous glucose production (k_{p1}) and basal insulin secretion (m_6), parameters which dictate the transition from healthy to prediabetic or diabetic states.[42] Sensitivity analyses of the DM model demonstrate that endogenous glucose production has a massive impact on sustained plasma glucose levels, whereas insulin-independent glucose utilization (V_{m0}) exerts a minor effect.[42] In the context of a diabetes application, tracking the intra-patient drift of k_{p1} over time allows the software to maintain long-term predictive accuracy, particularly during overnight fasting periods.[42]

5. Adaptive Algorithms: Real-Time Updates from Mid-Course Feedback

The core functional innovation requested for this application is the ability to ingest an actual blood glucose reading 30+ minutes after a meal and utilize that localized ground-truth data to instantly recalibrate the projected curve. Because physiological parameters—such as the insulin sensitivity index (S_I) and the time-to-maximum gastric emptying ($t_{max,G}$)—vary intra-individually from day to day by as much as 30% due to stress, sleep architecture, and exercise, static models inevitably drift into error.[34] Continuous feedback loops using optimal estimation and machine learning resolve this drift, forming the crux of the application's intelligence.[2]

5.1 Recursive Least Squares (RLS) Parameter Estimation

To mathematically update the prediction curve at $T + 30$ minutes, the application must implement Recursive Least Squares (RLS) parameter estimation. RLS is an adaptive filter algorithm utilized in system identification that recursively estimates the time-varying coefficients of an autoregressive moving average exogenous (ARMAX) time-series model as new data sequentially arrives.[2]

When the user inputs a 30-minute postprandial BG value, $y(t)$, the RLS algorithm attempts to find the parameter vector $\hat{\theta}$ that minimizes the weighted sum of squared errors between the physiological model's a priori prediction $\phi^T(i)\theta$ and the actual user measurement $y(i)$.[46] The cost function minimized by the application is defined as:

$$V(\theta, t) = \frac{1}{2} \sum_{i=1}^{t} \lambda^{t-i} (y(i) - \phi^T(i)\theta)^2$$ [47]

In this equation, λ represents the "forgetting factor" ($0 < \lambda \leq 1$), which dictates how much mathematical weight is given to recent versus historical data.[48] A λ value calibrated between 0.95 and 0.99 allows the algorithm to gracefully "forget" the patient's baseline insulin sensitivity from the previous week and heavily weight the immediate 30-minute reading.[22] This allows the app to instantly correct its projected trajectory for the remaining 3 to 4 hours of the meal absorption window.

5.2 Extended and Unscented Kalman Filtering (EKF/UKF)

While RLS efficiently updates model coefficients, Bayesian state-estimation techniques—specifically the Extended Kalman Filter (EKF) and the Unscented Kalman Filter (UKF)—are optimal for simultaneously updating both the internal *state* of the system (e.g., the current unobservable glucose levels in the gut and interstitial fluid) and the *parameters* (e.g., real-time insulin sensitivity).[45]

In a dual Kalman Filter framework implemented within the application, the workflow operates as follows:

1. **Prior Forecast (A Priori):** At the time of the meal ($T = 0$), the application utilizes the Hovorka/Bergman physiological model to project a forward glucose trajectory based on the meal's carbohydrate and FPU inputs.[36]
2. **Observation:** At $T = 30$, the user logs an actual BG value (or the app pulls it automatically via Bluetooth from a CGM).[2]
3. **Measurement Update (A Posteriori):** The Kalman Filter calculates the residual—the

difference between the 30-minute prediction and the actual 30-minute log. It multiplies this residual by the Kalman Gain, a mathematically derived matrix that balances the uncertainty of the physiological model against the inherent noise of the glucose sensor.[51]

4. **Trajectory Correction:** The filter corrects the internal state of the gastrointestinal compartment $R_a(t)$.[49] For example, if the 30-minute reading is significantly higher than projected, the EKF infers that gastric emptying was faster than modeled, increases the internal R_a state variable, and repaints a steeper, earlier peak on the user's interface.[50]

This approach, frequently deployed in data assimilation engines, infuses mechanistic models with real-time empirical data via Bayesian statistics, generating highly adaptable, individualized glucose forecasts that react dynamically to the patient's immediate physiological reality.[49]

5.3 Deep Learning and Bayesian Transfer Learning

For applications handling rich, continuous data streams from CGMs, traditional linear regression and compartmental models are increasingly augmented with Deep Learning architectures, particularly Long Short-Term Memory (LSTM) networks.[53] LSTMs are uniquely capable of capturing the non-linear, long-term dependencies of blood glucose dynamics, including the delayed effects of HFHP meals. When trained on robust clinical datasets, LSTMs and attention-based recurrent neural networks consistently achieve Root Mean Square Errors (RMSE) of under 20 mg/dL for 30-minute prediction horizons.[56]

However, deep learning models suffer from a "cold start" problem; they require vast amounts of individual data before they can generate accurate personalized predictions. To resolve this within the application, Bayesian Transfer Learning is utilized. The application's base model is pretrained on randomized controlled trial (RCT) data from a broad cohort.[3] When the patient inputs their first few 30-minute post-meal readings, the pretrained parameter distribution is computationally adjusted, transferred, and utilized to rapidly personalize the model to the specific patient, circumventing the need for months of initial data logging.[3]

6. Longitudinal Personalization: Learning Over Time

In addition to acute, intra-meal course corrections at 30 minutes, the application must fulfill the requirement to become "more accurate the more the patient uses the app." If a patient repeatedly overshoots their target blood glucose after eating a specific meal type, the foundational Insulin-to-Carbohydrate Ratio (ICR) or Insulin Sensitivity Factor (ISF) must be systematically altered. This is achieved through Run-to-Run (R2R) control algorithms and Reinforcement Learning.[4]

6.1 Run-to-Run (R2R) Control Algorithms

Run-to-Run (R2R) control, a concept adapted from chemical process engineering, treats each

day (or each specific meal type) as a discrete "run" in a batch process.[4] The application assesses the performance metric of the previous run (e.g., the 2-hour postprandial BG error) and algorithmically updates the dosing matrix for the subsequent run.[60]

The clinical heuristic underpinning R2R parameter adjustment is structured within the application logic as follows:

- If the 2-hour postprandial glucose consistently remains > 180 mg/dL (indicating under-dosing or severe insulin resistance), the app's algorithm decreases the ICR denominator (e.g., moving the ratio from 1:15 to 1:12 or 1:10), effectively recommending more insulin per carbohydrate gram for the next identical meal.[62]
- Conversely, if the postprandial trace indicates hypoglycemia (< 70 mg/dL), the algorithm acts conservatively, immediately increasing the ICR by 10-20% and reducing the Total Daily Dose (TDD) baseline to prioritize patient safety above all other metrics.[62]

By integrating R2R logic, the application transforms from a static mathematical calculator into a self-tuning decision support system that automatically tracks and adapts to variations in insulin sensitivity across circadian cycles, hormonal shifts, and long-term lifestyle changes.[64]

6.2 Deep Reinforcement Learning (DRL)

At the most advanced level of longitudinal learning, the application can deploy Deep Reinforcement Learning (DRL). DRL utilizes algorithms like Proximal Policy Optimization (PPO) or Twin Delayed Deep Deterministic Policy Gradients (TD3) to optimize insulin dosing policies.[66]

In this architecture, the DRL algorithm acts as an "agent" that receives a mathematical "reward" for keeping the forecasted blood glucose within the target Time in Range (TIR: 70-180 mg/dL) and a severe "penalty" for predicting or causing hypoglycemia.[68] Because reinforcement learning is inherently a trial-and-error process—which is dangerous in a clinical setting—the DRL agent is bounded by a dual mechanism of "proactive guidance + reactive correction".[66] A baseline Model Predictive Control (MPC) algorithm simulates the absolute physiological safety boundaries, effectively hard-coding limits that restrict the highly exploratory DRL agent from ever recommending catastrophically unsafe insulin doses.[68]

7. Current Clinical Guidelines (ADA & ISPAD 2024-2026)

To ensure the application is medically approved and aligned with standard-of-care practices, its internal logic must reflect the most recent clinical guidelines.

The **American Diabetes Association (ADA) Standards of Care (2025/2026)** heavily emphasize the utilization of Time in Range (TIR) metrics derived from CGM data. The clinical

target is to maintain BG levels between 70–180 mg/dL for $> 70\%$ of the day, while strictly limiting Time Below Range (TBR, < 70 mg/dL) to $< 4\%$, and level 2 TBR (< 54 mg/dL) to $< 1\%$.[70] The ADA formally recommends that individuals with Type 1 diabetes receive specific education on how to match mealtime insulin doses not just to carbohydrate intake, but explicitly to fat and protein intake.[72]

The **International Society for Pediatric and Adolescent Diabetes (ISPAD) 2024 Guidelines** reinforce this paradigm. ISPAD officially recognizes that fat and protein intake affects insulin requirements and validates the use of Fat-Protein Units (FPUs). The guidelines explicitly state that one FPU (100 kcal of fat or protein) requires the equivalent insulin of 10 grams of carbohydrates, delivered as an extended bolus to prevent delayed hyperglycemia.[74] The application's adherence to these specific consensus statements is critical for establishing its clinical validity during regulatory review.

8. FDA Regulatory Compliance: Software as a Medical Device (SaMD)

Developing a blood glucose projection and insulin recommendation application transitions the software entirely out of the realm of "general wellness" products and strictly into the jurisdiction of the FDA's regulatory framework for Software as a Medical Device (SaMD).[76] Because inaccurate glucose predictions or erroneous insulin dosing recommendations can lead to severe hypoglycemia, diabetic ketoacidosis, coma, or death, the software requires rigorous federal oversight.[78]

8.1 FDA Classification and Product Code QRX

Under the 21st Century Cures Act, clinical decision support (CDS) software that processes patient-specific physiological parameters to calculate dosages, generate predictive curves, or formulate specific treatment plans is legally defined as a medical device.[76] Specifically, software functioning as a "Continuous Glucose Monitor Informed Insulin Dose Calculator" falls under the regulation **21 CFR 862.1358** (Insulin Therapy Adjustment Device) and is assigned the **FDA Product Code QRX**.[81]

The FDA classifies these predictive applications as **Class II (Moderate Risk) devices**, which require a 510(k) premarket notification clearance prior to commercial distribution.[82] Achieving 510(k) clearance demands that the software developer prove the application is "substantially equivalent" in safety and efficacy to an already legally marketed predicate device.[84]

Recent successful predicate devices in this category include:

- **DreaMed Advisor Pro (DEN170043):** AI-based decision support that utilizes

retrospective data to recommend adjustments to basal rates, ICR, and ISF.[85]

- **Omnipod 5 SmartBolus Calculator (K203772):** Calculates suggested bolus doses based on carbohydrates, sensor glucose trends, and Insulin On Board.[87]
- **WellDoc BlueStar (K222888):** A digital health app cleared for adaptive insulin dosing recommendations.[88]
- **Dexcom Smart Basal (K252818):** Software designed to optimize basal insulin doses based on continuous monitoring data.[89]

8.2 Mandatory Special Controls (21 CFR 862.1358)

To avoid automatic designation as a Class III (High Risk) device—which would require a vastly more expensive and time-consuming Premarket Approval (PMA) process—the predictive software must adhere strictly to the "Special Controls" outlined in 21 CFR 862.1358.[83] The application's design, verification, and validation documentation submitted to the FDA must explicitly demonstrate compliance with the following mandates:

1. **Algorithmic Transparency and Data Inputs:** A complete mathematical description of the required data inputs must be provided. This includes the exact timeframe over which data is collected (e.g., 5-minute CGM intervals, 30-minute user-logged mid-course corrections) and the absolute minimum number of data points required for the algorithm to generate an accurate glycemic prediction without defaulting to an error state.[91]
2. **Output Justification:** A detailed functional description of how the adaptive algorithms (RLS, Kalman Filters, Neural Networks) generate the device outputs, predictive curves, and insulin therapy adjustment recommendations.[91]
3. **Clinical Validity:** Robust clinical trial data demonstrating the accuracy of the glucose projections. Performance is heavily evaluated using standard metrics such as Root Mean Square Error (RMSE) and the Surveillance Error Grid analysis. For FDA clearance, an overwhelming majority (>95%) of algorithmic predictions must fall within clinically safe zones (Zones A and B) to prove that modeling errors do not result in catastrophic dosing recommendations.[56]
4. **Mitigation of Dosing Errors (Safety Constraints):** The software architecture must include an appropriate, hard-coded mitigation strategy to minimize extreme recommendation errors.[91] If the DRL or RLS algorithm calculates an unusually aggressive insulin dose due to a false 30-minute blood glucose spike (e.g., induced by sensor compression noise rather than physiological glucose), the application must have fail-safes (such as Maximum Insulin limits or algorithmic suspension) to mitigate the risk of severe hypoglycemia.[68]
5. **Cybersecurity and Data Transmission:** A detailed strategy to ensure secure and reliable data transmission via Bluetooth or cloud APIs, including data integrity checks, accuracy protocols, and advanced security measures to prevent data corruption or adversarial hacking, which could lead to inappropriate and life-threatening treatment recommendations.[83]

8.3 Insulin On Board (IOB) and Alert Mechanisms

To conform to FDA safety paradigms and avoid the critical danger of "insulin stacking," the application must natively and continuously track "Insulin On Board" (IOB), also known as active insulin.[93] Because the algorithm recalculates predictions and potential correction doses at 30-minute intervals, it must accurately deduct the residual metabolic action of any previously administered insulin doses from any newly recommended correction boluses.[93]

Furthermore, the FDA requires strict human factors engineering for application alerts. Missing an algorithmic alert for predicted severe hypoglycemia can be fatal. The FDA recently issued safety communications stressing that software must ensure robust notification architectures that can override smartphone operating system updates, "Do Not Disturb" modes, and deep sleep states to guarantee the audible delivery of critical predictive warnings to the patient.[79]

9. Conclusion

Engineering an advanced diabetic application capable of forecasting postprandial blood glucose and adaptively refining its predictions from a 30-minute post-meal reading requires a highly sophisticated synthesis of nutritional science, physiological mathematical modeling, and machine learning. Standard carbohydrate counting is insufficient for modern automated diabetes management. The system must natively deploy mixed-meal heuristics—such as the Pankowska Equation's Fat-Protein Units or the Sieradzki Equation's extended dosing proportions—to mathematically account for the delayed glycemic absorption patterns induced by dietary fats and proteins.

These heuristics serve as initialization inputs to physiological differential architectures, such as the Hovorka Model, which map the complex transition of glucose from the gastrointestinal tract to the blood plasma. The defining innovation of the proposed application lies in its adaptive feedback loop. By applying Recursive Least Squares (RLS) parameter estimation or Unscented Kalman Filtering, the application can ingest the 30-minute postprandial logged value, calculate the residual error of the initial forecast, and instantaneously recalculate the patient’s real-time insulin sensitivity and gastric absorption rates to output a highly accurate short-term curve. Over time, Run-to-Run (R2R) control algorithms and bounded Deep Reinforcement Learning ensure the longitudinal personalization of the patient's core metabolic ratios, making the app inherently smarter with continued use.

Finally, the deployment of this sophisticated, adaptive intelligence is strictly governed by medical device regulations. To achieve market viability, the application must fulfill the rigorous design validation, clinical transparency, and fail-safe safety constraints mandated by FDA 21 CFR 862.1358 Special Controls (Product Code QRX). By successfully harmonizing robust mathematical modeling with rigorous regulatory compliance and human-centric safety boundaries, the application can safely transition from theoretical forecasting to an actionable,

life-saving clinical decision support tool for individuals managing diabetes.

Works cited

1. Factoring fat, protein and carbs - Children with Diabetes, accessed February 26, 2026, https://media.childrenwithdiabetes.com/presentations/Marissa_-_FFL_2022_pro__fat_Groppo.pdf
2. Adaptive Personalized Prior-Knowledge-Informed Model Predictive Control for Type 1 Diabetes - PMC, accessed February 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC9730892/
3. Optimizing postprandial glucose prediction through integration of diet and exercise: Leveraging transfer learning with imbalanced patient data - PMC, accessed February 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC11293722/
4. A Run-to-Run Control Strategy to Adjust Basal Insulin Infusion Rates in Type 1 Diabetes - PMC, accessed February 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC2516944/
5. Insulin Correction Dose Calculator - Perinatology.com, accessed February 26, 2026, https://www.perinatology.com/calculators/Insulin%20Correction%20Dose.htm
6. Both Dietary Protein and Fat Increase Postprandial Glucose Excursions in Children With Type 1 Diabetes, and the Effect Is Additive - PMC, accessed February 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC3836096/
7. My Carbohydrate Guide - Alabama Department of Public Health (ADPH), accessed February 26, 2026, https://www.alabamapublichealth.gov/diabetes/assets/carbguide.pdf
8. Carbohydrate Counting Handbook, accessed February 26, 2026, https://www.chkd.org/media/wxydqjty/carbohydrate-counting-handbook-chkd-diabetes-program.pdf
9. Impact of Diabetes-Specific Nutritional Formulas versus Oatmeal on Postprandial Glucose, Insulin, GLP-1 and Postprandial Lipidemia - MDPI, accessed February 26, 2026, https://www.mdpi.com/2072-6643/8/7/443
10. Carbohydrate Counting | Alberta Health Services, accessed February 26, 2026, https://www.albertahealthservices.ca/assets/info/nutrition/if-nfs-carbohydrate-counting.pdf
11. Bolusing for Fat and Protein - Children with Diabetes, accessed February 26, 2026, https://childrenwithdiabetes.com/clinical-director/bolusing-for-fat-and-protein/
12. The effect of the modified fat-protein unit algorithm compared with that of carbohydrate counting on postprandial glucose in adults with type-1 diabetes when consuming meals with differing macronutrient compositions: a randomized crossover trial - PMC, accessed February 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC10580506/
13. Protein and Diabetes: What to Eat & How Much You Need | Dexcom, accessed February 26, 2026,

https://www.dexcom.com/en-us/all-access/clinical-corner/protein-and-diabetes
14. Recent consumption of a caffeine-containing beverage and serum biomarkers of cardiometabolic function in the UK Biobank - PMC, accessed February 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC8093320/
15. Caffeine: Does it affect blood sugar? - Mayo Clinic, accessed February 26, 2026, https://www.mayoclinic.org/diseases-conditions/type-2-diabetes/expert-answers/blood-sugar/faq-20057941
16. CHAPTER 3: CALCULATION OF THE ENERGY CONTENT OF FOODS - ENERGY CONVERSION FACTORS - Food and Agriculture Organization of the United Nations, accessed February 26, 2026, https://www.fao.org/4/y5022e/y5022e04.htm
17. A guide to carbohydrate counting and dose adjustment - Gilberdyke Health Centre, accessed February 26, 2026, https://www.gilberdykehealthcentre.nhs.uk/wp-content/uploads/2025/04/A-guide-to-carbohydrate-counting-and-dose-adjustment.pdf
18. EMPOWERING PEOPLE WITH TYPE 1 DIABETES IN PHYSICAL ACTIVITY - POLITesi - Politecnico di Milano, accessed February 26, 2026, https://www.politesi.polimi.it/retrieve/8ff6b3ee-e37b-4eb8-89ed-0aec937cbb2e/2024_12_Bonaldo.pdf
19. Full text of "The Journal Of Pediatrics(45)" - Internet Archive, accessed February 26, 2026, https://archive.org/stream/in.ernet.dli.2015.67119/2015.67119.The-Journal-Of-Pediatrics45_djvu.txt
20. Counting fat and protein: A dietitian's perspective - DiabetesontheNet, accessed February 26, 2026, https://diabetesonthenet.com/diabetes-care-children-young-people/counting-fat-and-protein-a-dietitians-perspective/
21. The Impact of Two Different Insulin Dose Calculation Methods on Postprandial Glycemia After a Mixed Meal in Children with Type 1 Diabetes - PMC, accessed February 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC12567265/
22. Fat and Protein Insulin Calculator — JUICEBOXPODCAST.com, accessed February 26, 2026, https://www.juiceboxpodcast.com/warcal
23. Warsaw Method Calculator, accessed February 26, 2026, https://www.omnicalculator.com/health/warsaw-method
24. Protein, Fat and BGs - T1D Academy, accessed February 26, 2026, https://www.t1dacademy.ca/post/protein-fat-and-blood-sugars
25. Tame Post-Meal Spikes: Fat & Protein - Waltzing the Dragon, accessed February 26, 2026, https://waltzingthedragon.ca/diabetes/nutrition-excercise/reduce-post-meal-spikes-caused-by-fat-and-protein/
26. Insulin for Protein and Fat - Diabetes Educators Calgary, accessed February 26, 2026, https://diabeteseducatorscalgary.ca/medications/insulin/insulin-for-protein-and-fat.html
27. Tim Jones - The Kids Research Institute Australia, accessed February 26, 2026,

https://www.thekids.org.au/contact-us/our-people/j/tim-jones/
28. Glucose excursions (mmol/L) at 30-minute intervals by continuous... - ResearchGate, accessed February 26, 2026, https://www.researchgate.net/figure/Glucose-excursions-mmol-L-at-30-minute-intervals-by-continuous-glucose-monitoring_fig1_374767525
29. (PDF) The Impact of Two Different Insulin Dose Calculation Methods on Postprandial Glycemia After a Mixed Meal in Children with Type 1 Diabetes: A Randomized Study - ResearchGate, accessed February 26, 2026, https://www.researchgate.net/publication/396732779_The_Impact_of_Two_Different_Insulin_Dose_Calculation_Methods_on_Postprandial_Glycemia_After_a_Mixed_Meal_in_Children_with_Type_1_Diabetes_A_Randomized_Study
30. Nutrients, Volume 17, Issue 20 (October-2 2025) – 130 articles - MDPI, accessed February 26, 2026, https://www.mdpi.com/2072-6643/17/20
31. Clinical application of the food insulin index for mealtime insulin dosing in adults with type 1 diabetes: A randomized controlled trial - Macquarie University, accessed February 26, 2026, https://researchers.mq.edu.au/en/publications/clinical-application-of-the-food-insulin-index-for-mealtime-insul/
32. Clinical Application of the Food Insulin Index for Mealtime Insulin Dosing in Adults with Type 1 Diabetes: A Randomized Controlled Trial - PubMed, accessed February 26, 2026, https://pubmed.ncbi.nlm.nih.gov/26840067/
33. Novel food insulin index algorithm improved glycemic control | MDedge - The Hospitalist, accessed February 26, 2026, https://blogs.the-hospitalist.org/content/novel-food-insulin-index-algorithm-improved-glycemic-control
34. Long-Term Glucose Forecasting Using a Physiological Model and Deconvolution of the Continuous Glucose Monitoring Signal - MDPI, accessed February 26, 2026, https://www.mdpi.com/1424-8220/19/19/4338
35. MATHEMATICAL ANALYSIS OF A MODEL FOR GLUCOSE REGULATION Kimberly Fessel and Jeffrey B. Gaither Julie K. Bower Trudy Gaillard and - AIMS Press, accessed February 26, 2026, https://www.aimspress.com/aimspress-data/mbe/2016/1/PDF/1551-0018_2016_1_83.pdf
36. Dynamic Modeling of Exercise Effects on Plasma Glucose and Insulin Levels - PMC - NIH, accessed February 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC2769581/
37. A Mathematical Modeling Approach to Estimate Blood Glucose Behavior in Individuals with Prediabetes - MDPI, accessed February 26, 2026, https://www.mdpi.com/2673-4540/6/11/123
38. Minimally-Invasive and Efficient Method to Accurately Fit the Bergman Minimal Model to Diabetes Type 2 - PMC, accessed February 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC9124285/
39. Mathematical meal models for simulation of human metabolism - arXiv.org, accessed February 26, 2026, https://arxiv.org/pdf/2307.16444
40. Model Predictive Control Using the Improved Hovorka Model for the Regulation

of Blood Glucose Levels in Type 1 Diabetes - MDPI, accessed February 26, 2026, https://www.mdpi.com/2079-9292/14/23/4585
41. In-Silico Works Using an Improved Hovorka Equations Model and Clinical Works on the Control of Blood Glucose Levels in People With Type 1 Diabetes: Comparison Study - JMIRx Bio, accessed February 26, 2026, https://xbio.jmir.org/2024/1/e43662
42. Full article: Modeling glucose–insulin dynamics to evaluate healthy and diseased population characteristics in type 2 diabetes mellitus - Taylor & Francis, accessed February 26, 2026, https://www.tandfonline.com/doi/full/10.1080/29937574.2024.2431821
43. Experimental Evaluation of a Recursive Model Identification Technique for Type 1 Diabetes, accessed February 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC2769906/
44. Predicting Postprandial Glycemic Responses With Limited Data in Type 1 and Type 2 Diabetes - PMC, accessed February 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC11883769/
45. Personalized State-space Modeling of Glucose Dynamics for Type 1 Diabetes Using Continuously Monitored Glucose, Insulin Dose, and Meal Intake: An Extended Kalman Filter Approach - PMC, accessed February 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC4455398/
46. Adaptive System Identification for Estimating Future Glucose Concentrations and Hypoglycemia Alarms - PMC, accessed February 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC3409594/
47. A Personalized Week-to-Week Updating Algorithm to Improve Continuous Glucose Monitoring Performance - PMC, accessed February 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC5951058/
48. Forecasting of Glucose Levels and Hypoglycemic Events: Head-to-Head Comparison of Linear and Nonlinear Data-Driven Algorithms Based on Continuous Glucose Monitoring Data Only - MDPI, accessed February 26, 2026, https://www.mdpi.com/1424-8220/21/5/1647
49. Personalized glucose forecasting for type 2 diabetes using data assimilation, accessed February 26, 2026, https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1005232
50. (PDF) Personalized State-space Modeling of Glucose Dynamics for Type 1 Diabetes Using Continuously Monitored Glucose, Insulin Dose, and Meal Intake: An Extended Kalman Filter Approach - ResearchGate, accessed February 26, 2026, https://www.researchgate.net/publication/262787190_Personalized_State-space_Modeling_of_Glucose_Dynamics_for_Type_1_Diabetes_Using_Continuously_Monitored_Glucose_Insulin_Dose_and_Meal_Intake_An_Extended_Kalman_Filter_Approach
51. Blood Glucose Prediction Algorithms for Hypoglycemic and/or Hyperglycemic Alerts - SciSpace, accessed February 26, 2026, https://scispace.com/pdf/blood-glucose-prediction-algorithms-for-hypoglycemic-and-or-4vju5feuid.pdf

52. Bayesian hierarchical approach to estimate insulin sensitivity by minimal model - PubMed, accessed February 26, 2026, https://pubmed.ncbi.nlm.nih.gov/12790795/
53. Cyber–Physical–Human Systems in Precision Medicine: Advances in Artificial Pancreas for Treatment of Diabetes - PMC, accessed February 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC12768518/
54. Generalization of a Deep Learning Model for Continuous Glucose Monitoring–Based Hypoglycemia Prediction: Algorithm Development and Validation Study - JMIR Medical Informatics, accessed February 26, 2026, https://medinform.jmir.org/2024/1/e56909/
55. Personalized Blood Glucose Forecasting From Limited CGM Data Using Incrementally Retrained LSTM - Stevens Institute of Technology, accessed February 26, 2026, https://researchwith.stevens.edu/en/publications/personalized-blood-glucose-forecasting-from-limited-cgm-data-usin/
56. Machine learning-based glucose prediction with use of continuous glucose and physical activity monitoring data: The Maastricht Study - PMC, accessed February 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC8224858/
57. Personalized Blood Glucose Prediction for Type 1 Diabetes Using Evidential Deep Learning and Meta-Learning - PubMed, accessed February 26, 2026, https://pubmed.ncbi.nlm.nih.gov/35776825/
58. Personalized blood glucose prediction in type 1 diabetes using meta-learning with bidirectional long short term memory-transformer hybrid model - PMC, accessed February 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC12368048/
59. Optimizing postprandial glucose prediction through integration of diet and exercise: Leveraging transfer learning with imbalanced patient data - PubMed, accessed February 26, 2026, https://pubmed.ncbi.nlm.nih.gov/39088422/
60. Clinical Update on Optimal Prandial Insulin Dosing Using a Refined Run-to-Run Control Algorithm - PMC, accessed February 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC2769876/
61. Prandial insulin dosing using run-to-run control: application of clinical data and medical expertise to define a suitable performance metric - PubMed, accessed February 26, 2026, https://pubmed.ncbi.nlm.nih.gov/17303792/
62. What adjustment should be made to the insulin-to-carbohydrate (I:C) ratio for a patient with diabetes, likely type 1 or insulin-treated type 2, who has a blood glucose level of hyperglycemia after a carb ratio of 1 unit per 15 grams of carbohydrates? - Dr.Oracle, accessed February 26, 2026, https://www.droracle.ai/articles/805173/what-adjustment-should-be-made-to-the-insulin-to-carbohydrate-ic
63. Guidelines for Optimal Bolus Calculator Settings in Adults - PMC, accessed February 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC3045234/
64. Insulin Sensitivity Index-Based Optimization of Insulin to Carbohydrate Ratio: In Silico Study Shows Efficacious Protection Against Hypoglycemic Events Caused by Suboptimal Therapy - PMC, accessed February 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC5771547/
65. Adaptive Algorithms for Personalized Diabetes Treatment | Request PDF -

ResearchGate, accessed February 26, 2026, https://www.researchgate.net/publication/300822145_Adaptive_Algorithms_for_Personalized_Diabetes_Treatment
66. A safe-enhanced fully closed-loop artificial pancreas controller based on deep reinforcement learning | PLOS One - Research journals, accessed February 26, 2026, https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0317662
67. Blood Glucose Level Prediction in Type 1 Diabetes Using Machine Learning - arXiv, accessed February 26, 2026, https://arxiv.org/html/2502.00065v1
68. A Hybrid Closed-Loop Blood Glucose Control Algorithm with a Safety Limiter Based on Deep Reinforcement Learning and Model Predictive Control - MDPI, accessed February 26, 2026, https://www.mdpi.com/2079-6374/16/1/47
69. Reinforcement Learning: A Paradigm Shift in Personalized Blood Glucose Management for Diabetes - PMC, accessed February 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC11429314/
70. Distinct Roles of Common Genetic Variants and Their Contributions to Diabetes: MODY and Uncontrolled T2DM - MDPI, accessed February 26, 2026, https://www.mdpi.com/2218-273X/15/3/414
71. Consensus statement on novel glucose-related metrics obtained through advanced medical devices: English version - PMC, accessed February 26, 2026, https://pmc.ncbi.nlm.nih.gov/articles/PMC12532493/
72. 9. Pharmacologic Approaches to Glycemic Treatment: Standards of Care in Diabetes—2025, accessed February 26, 2026, https://diabetesjournals.org/care/article/48/Supplement_1/S181/157569/9-Pharmacologic-Approaches-to-Glycemic-Treatment
73. Show HN: The App I Built to Help Manage My Diabetes - Hacker News, accessed February 26, 2026, https://news.ycombinator.com/item?id=42168491
74. International Society for Pediatric and Adolescent Diabetes Clinical Practice Consensus Guidelines 2024: Insulin and Adjunctive, accessed February 26, 2026, https://ima-contentfiles.s3.amazonaws.com/ICPCG-Chapter-9-2024.pdf
75. ISPAD Clinical Practice Consensus Guidelines 2024: Glycemic Targets - ResearchGate, accessed February 26, 2026, https://www.researchgate.net/publication/387237797_ISPAD_Clinical_Practice_Consensus_Guidelines_2024_Glycemic_Targets
76. Mobile Health App Interactive Tool | Federal Trade Commission, accessed February 26, 2026, https://www.ftc.gov/business-guidance/resources/mobile-health-apps-interactive-tool
77. FDA SaMD (Software as Medical Device) Compliance Guide 2026, accessed February 26, 2026, https://www.tactionsoft.com/blog/fda-samd-software-as-medical-device-compliance-guide/
78. Do Not Use Smartwatches or Smart Rings to Measure Blood Glucose Levels: FDA Safety Communication - Food and Drug Administration, accessed February 26, 2026, https://www.fda.gov/medical-devices/safety-communications/do-not-use-smart

watches-or-smart-rings-measure-blood-glucose-levels-fda-safety-communication
79. FDA Alerts Patients to Regularly Check Diabetes-Related Smartphone Device Alert Settings, Especially Following Phone Hardware or Software Changes: FDA Safety Communication, accessed February 26, 2026, https://www.fda.gov/medical-devices/safety-communications/fda-alerts-patients-regularly-check-diabetes-related-smartphone-device-alert-settings-especially
80. Clinical Decision Support Software Frequently Asked Questions (FAQs) | FDA, accessed February 26, 2026, https://www.fda.gov/medical-devices/software-medical-device-samd/clinical-decision-support-software-frequently-asked-questions-faqs
81. Product Classification - Food and Drug Administration, accessed February 26, 2026, https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpcd/classification.cfm?start_search=1&productcode=QRX
82. 21 CFR § 862.1358 - Insulin therapy adjustment device. - Law.Cornell.Edu, accessed February 26, 2026, https://www.law.cornell.edu/cfr/text/21/862.1358
83. Medical Devices; Clinical Chemistry and Clinical Toxicology Devices; Classification of the Insulin Therapy Adjustment Device - Federal Register, accessed February 26, 2026, https://www.federalregister.gov/documents/2018/11/01/2018-23912/medical-devices-clinical-chemistry-and-clinical-toxicology-devices-classification-of-the-insulin
84. FDA Clears First Device to Enable Automated Insulin Dosing for Individuals with Type 2 Diabetes, accessed February 26, 2026, https://www.fda.gov/news-events/press-announcements/fda-clears-first-device-enable-automated-insulin-dosing-individuals-type-2-diabetes
85. Insulin - accessdata.fda.gov, accessed February 26, 2026, https://www.accessdata.fda.gov/cdrh_docs/pdf17/DEN170043.pdf
86. The DreaMed Advisor Pro workflow. | Download Scientific Diagram - ResearchGate, accessed February 26, 2026, https://www.researchgate.net/figure/The-DreaMed-Advisor-Pro-workflow_fig1_346406982
87. 510(k) SUBSTANTIAL EQUIVALENCE DETERMINATION DECISION SUMMARY - accessdata.fda.gov, accessed February 26, 2026, https://www.accessdata.fda.gov/cdrh_docs/reviews/K203772.pdf
88. A 510(k) Number K222888 B Applicant Well - accessdata.fda.gov, accessed February 26, 2026, https://www.accessdata.fda.gov/cdrh_docs/reviews/K222888.pdf
89. FDA clears Dexcom's Type 2 diabetes software for optimizing basal insulin doses, accessed February 26, 2026, https://www.fiercebiotech.com/medtech/fda-clears-dexcoms-type-2-diabetes-software-optimizing-basal-insulin-doses
90. Dexcom, Inc. November 18, 2025 Dhara Sanjaykumar Desai Senior Regulatory Affairs Specialist 6340 Sequence Drive San Diego, Calif, accessed February 26,

2026, https://www.accessdata.fda.gov/cdrh_docs/pdf25/K252818.pdf
91. 21 CFR 862.1358 -- Insulin therapy adjustment device. - eCFR, accessed February 26, 2026, https://www.ecfr.gov/current/title-21/chapter-I/subchapter-H/part-862/subpart-B/section-862.1358
92. Personalized blood glucose prediction: A hybrid approach using grammatical evolution and physiological models | PLOS One - Research journals, accessed February 26, 2026, https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0187754
93. Guide to Successful Pumping - Tandem Diabetes Care, accessed February 26, 2026, https://www.tandemdiabetes.com/docs/default-source/general-guides/ml-1002246_a_a_guide_to_successful_pumping_webe24f699775426a79a519ff0b00a9fd39e34f699775426a79a519ff0b00a9fd39.pdf?sfvrsn=a8553bd7_70
94. (PDF) Guidelines for Optimal Bolus Calculator Settings in Adults - ResearchGate, accessed February 26, 2026, https://www.researchgate.net/publication/49819728_Guidelines_for_Optimal_Bolus_Calculator_Settings_in_Adults